Overview: Basics of REMs.

Boschi, M.

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Synthetic Data: Academics moving in US states.

Preliminarities

More details about the preliminaries can be found in the document: `01-simulation-study-implementation`.

```
source("script/states.R")
```

```
##   |                                                                      |
```

```
source("script/input_distance.R")
source("script/true_coefficients.R")
```

```
n_sim = 1
```

```
load(file="output/running_example_data.RData")
load(file="output/running_example_gam.fits.RData")
load(file="output/running_example_shifted_data.RData")
load(file="output/running_example_gam.fits_shifted.RData")
```

```
remdat <- remdats_nocox[[1]][[1]]
last_visited <- remdats_nocox[[1]][[2]]
shifted_remdat <- shifted_rems[[1]]
```

```
# check an event whose reciprocal is close in time
unlist(sapply(which(remdat$rec==1),
              function(x){
                if(min(abs(which(remdat$s==remdat[x,"r"] &
                                 remdat$r==remdat[x,"s"]) - x) < 10)){
                  return(x)
                }
              }))
```

```
## [1] 237 702 805
```

```
x1 <- 795
x2 <- 805
```

```
which(remdat$s==remdat[x2,"r"] & remdat$r==remdat[x2,"s"])
```

```
## [1] 796
```

```
remdat[c(795:805),]
```

```
##     index      tms  s  r event.id weekday  distance rec non.s non.r
## 795   795 381.2898 35 22     1211       1 4.7311974   0    22    31
## 796   796 381.3196 29 49     2717       1 2.2117898   0    48     3
```

```
## 797   797 381.8545 25 46     2545       1 0.6757388   1    32    36
## 798   798 382.9113 11 41     2251       1 4.6563958   0    49    49
## 799   799 383.1094  3  8      395       1 3.9308407   0    54    23
## 800   800 383.2879 40 24     1328       1 2.4699069   0    33    11
## 801   801 383.4272 26 44     2434       1 3.9268685   0    31    31
## 802   802 383.4998 39 54     3007       1 4.1586189   0     8    31
## 803   803 383.9824 26 41     2266       1 3.9268685   0     3    24
## 804   804 384.3754 31 55     3055       1 2.1353788   0    21    34
## 805   805 384.6348 49 29     1617       1 2.5410820   1    15    51
##     non.event.id non.distance non.rec delta.dist delta.rec one
## 795         1702     2.774312       0  1.9568854         0   1
## 796          160     0.000000       0  2.2117898         0   1
## 797         1992     2.139497       2 -1.4637582        -1   1
## 798         2737     3.099062       0  1.5573339         0   1
## 799         1286     3.545699       0  0.3851419         0   1
## 800          593     0.000000       1  2.4699069        -1   1
## 801         1711     2.806817       0  1.1200512         0   1
## 802         1688     0.000000       1  4.1586189        -1   1
## 803         1291     4.673678       0 -0.7468093         0   1
## 804         1869     1.976373       0  0.1590056         0   1
## 805         2815     1.640116       0  0.9009661         1   1
```

```r
# 26 repeats twice as sender
# check the previous occurrence
which(remdat$s==26)
```

```
##  [1]  66  92 320 361 365 397 439 449 520 536 604 661 720 801 803 837 927 934 944
```

```r
# select the closest to selected events
x3 <- 720
```

Table 1: Case-Control Dataset

```r
base_date <- as.Date("2023-01-01")
```

```r
remdat_display <- remdat[c(x3,x1:x2),c("tms",
                                       "s", "r",
                                       "non.s", "non.r")]
remdat_display <- remdat_display %>%
  mutate(tms = format(base_date + floor(tms), "%d.%m.%Y"))
# remdat_display$tms <- round(remdat_display$tms, 2)
remdat_display$s <- states[remdat_display$s]
remdat_display$r <- states[remdat_display$r]
remdat_display$non.s <- states[remdat_display$non.s]
remdat_display$non.r <- states[remdat_display$non.r]

remdat_display <- rbind(rep("...", ncol(remdat_display)),
                        remdat_display[1,],
                        rep("...", ncol(remdat_display)),
                        remdat_display[-1,],
                        rep("...", ncol(remdat_display)))

colnames(remdat_display) <- c("Time",
                              "Origin (Event)", "Destination (Event)",
                              "Origin (Non-Event)", "Destination (Non-Event)")
```

```r
rownames(remdat_display) <- NULL
(remdat_display <- rbind(remdat_display,
                         rep(" ", ncol(remdat_display))))
```

```
##          Time Origin (Event) Destination (Event)
## 1         ...            ...                 ...
## 2  21.12.2023       New York               Texas
## 3         ...            ...                 ...
## 4  17.01.2024           Iowa             Indiana
## 5  17.01.2024        Vermont          Washington
## 6  17.01.2024       Colorado        Rhode Island
## 7  18.01.2024  West Virginia               Texas
## 8  19.01.2024       Maryland                Guam
## 9  19.01.2024        Arizona            Nebraska
## 10 19.01.2024       New York              Hawaii
## 11 19.01.2024    Connecticut           Wisconsin
## 12 19.01.2024       New York               Texas
## 13 20.01.2024        Florida      North Carolina
## 14 20.01.2024     Washington             Vermont
## 15        ...            ...                 ...
## 16
##                              Origin (Non-Event) Destination (Non-Event)
## 1                                           ...                     ...
## 2                                         Texas                Virginia
## 3                                           ...                     ...
## 4                                       Indiana                 Florida
## 5                                South Carolina                Maryland
## 6                                        Nevada                 Alabama
## 7                                    Washington              Washington
## 8                                     Wisconsin              New Jersey
## 9  Commonwealth of the Northern Mariana Islands           West Virginia
## 10                                      Florida                 Florida
## 11                                         Guam                 Florida
## 12                                     Maryland                Nebraska
## 13                                     Oklahoma               Tennessee
## 14                                    Minnesota               Louisiana
## 15                                          ...                     ...
## 16
```

```r
ft <- flextable(remdat_display) %>%
  set_table_properties(layout = "fixed", width = 1) %>%
  width(j = c(2:5), width = 2) %>%
  fontsize(size = 9, part = "all")
save_as_image(ft, path="pictures/overview/overview_case_control.png")
```

```
## [1] "pictures/overview/overview_case_control.png"
```

Table 2: Covariate Computation

```r
remdat_display <- remdat[c(x3,x1:x2),c("tms",
                                 "s", "r",
                                 "distance", "rec", "weekday")]
remdat_display <- remdat_display %>%
  mutate(tms = format(base_date + floor(tms), "%d.%m.%Y"))
```

```r
remdat_display$s <- states[remdat_display$s]
remdat_display$r <- states[remdat_display$r]
remdat_display$distance <- round(remdat_display$distance, 2)
remdat_display <- rbind(rep("...", ncol(remdat_display)),
                        remdat_display[1,],
                        rep("...", ncol(remdat_display)),
                        remdat_display[-1,],
                        rep("...", ncol(remdat_display)))
colnames(remdat_display) <- c("Time",
                              "Origin (Event)", "Destination (Event)",
                              "Distance", "Reciprocity", "Weekday (Event)")
rownames(remdat_display) <- NULL
remdat_display <- rbind(remdat_display,
                        rep(" ", ncol(remdat_display)))
remdat_display
```

```
##          Time Origin (Event) Destination (Event) Distance Reciprocity
## 1         ...            ...                 ...      ...         ...
## 2  21.12.2023       New York               Texas     4.66           0
## 3         ...            ...                 ...      ...         ...
## 4  17.01.2024           Iowa             Indiana     4.73           0
## 5  17.01.2024        Vermont          Washington     2.21           0
## 6  17.01.2024       Colorado        Rhode Island     0.68           1
## 7  18.01.2024  West Virginia               Texas     4.66           0
## 8  19.01.2024       Maryland                Guam     3.93           0
## 9  19.01.2024        Arizona            Nebraska     2.47           0
## 10 19.01.2024       New York              Hawaii     3.93           0
## 11 19.01.2024    Connecticut           Wisconsin     4.16           0
## 12 19.01.2024       New York               Texas     3.93           0
## 13 20.01.2024        Florida      North Carolina     2.14           0
## 14 20.01.2024     Washington             Vermont     2.54           1
## 15        ...            ...                 ...      ...         ...
## 16
##    Weekday (Event)
## 1              ...
## 2                1
## 3              ...
## 4                1
## 5                1
## 6                1
## 7                1
## 8                1
## 9                1
## 10               1
## 11               1
## 12               1
## 13               1
## 14               1
## 15             ...
## 16
```

```r
dist_matrix["Texas","Hawaii"]
```

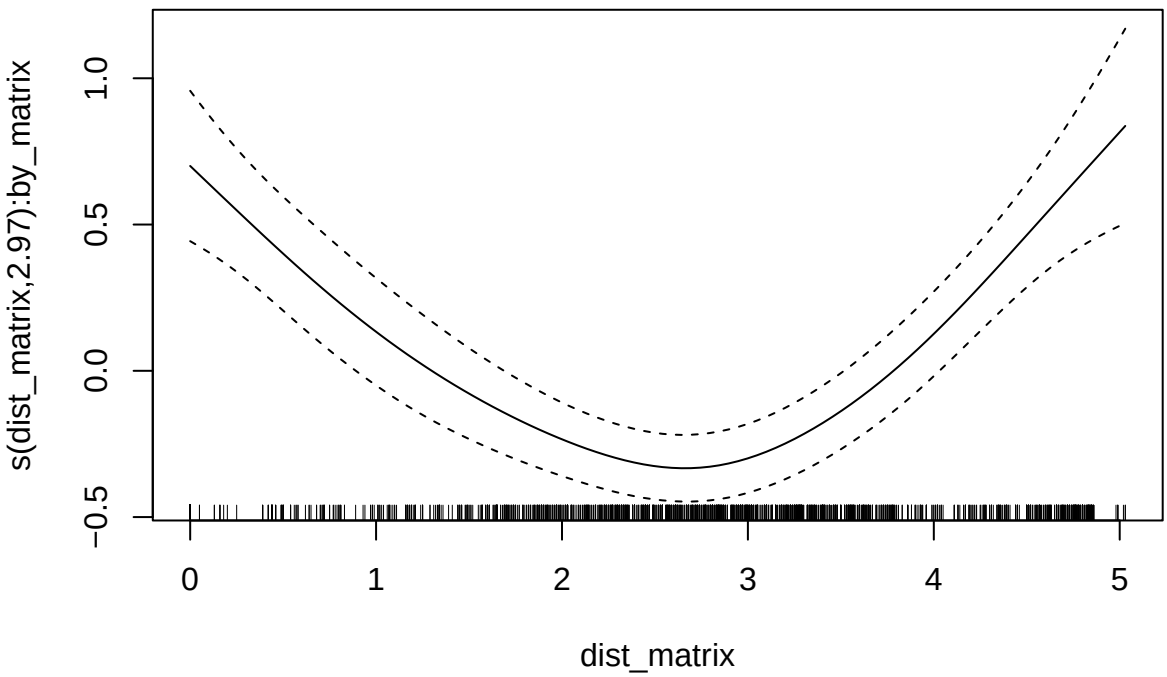
```
## [1] 3.926869
```

```r
ft <- flextable(remdat_display) %>%
  set_table_properties(layout = "fixed", width = 1) %>%
  width(j = c(2:5), width = 2) %>%
  fontsize(size = 9, part = "all")%>%
  bg(i = 5, j = 3, bg = "#FFE4E1")%>%
  bg(i = 5, j = 2, bg = "#FFE4E1")%>%
  bg(i = 14, j = 5, bg = "#FFE4E1")%>%
  bg(i = 14, j = 2, bg = "#FFE4E1")%>%
  bg(i = 14, j = 3, bg = "#FFE4E1")%>%

  bg(i = 2, j = 2, bg = "#ADD8E6")%>%
  bg(i = 10, j = 2, bg = "#ADD8E6")%>%
  bg(i = 12, j = 2, bg = "#ADD8E6")%>%

  bg(i = 2, j = 3, bg = "#FFFACD")%>%

  bg(i = 10, j = 3, bg = "#FFFACD")%>%
  bg(i = 10, j = 4, bg = "#FFFACD")%>%

  bg(i = 12, j = 3, bg = "#FFFACD")%>%
  bg(i = 12, j = 4, bg = "#FFFACD")
save_as_image(ft, path="pictures/overview/overview_covariate_computation.png")
```

```
## [1] "pictures/overview/overview_covariate_computation.png"
```

Table 3: Model Fitting

```r
rem_data_display <- remdat[,c("tms", "rec", "non.rec", "delta.rec",
                                  "distance", "non.distance")]
rem_data_display <- rem_data_display %>%
  mutate(tms = format(base_date + floor(tms), "%d.%m.%Y"))
rem_data_display$distance <- round(rem_data_display$distance, 2)
rem_data_display$non.distance <- round(rem_data_display$non.distance, 2)
rem_data_display$id_ev <- 1
rem_data_display$id_non_ev <- -1
rem_data_display$y <- 1
colnames(rem_data_display) <- c("time",
                                "rec_ev", "rec_non_ev", "delta.rec",
                                "dist_ev", "dist_non_ev",
                                "id_ev", "id_non_ev", "y")
rownames(rem_data_display) <- NULL
ncc_data <- rem_data_display
library(mgcv)
dist_matrix <- cbind(ncc_data$dist_ev, ncc_data$dist_non_ev)
by_matrix <- cbind(ncc_data$id_ev, ncc_data$id_non_ev)
gam_fit <- gam(y ~ -1 + s(dist_matrix, by=by_matrix) + delta.rec,
               family="binomial"(link = "logit"), data = ncc_data)
plot(gam_fit)
```




```
rem_data_display <- rem_data_display[c(x3,x1:x2),]
rem_data_display <- rbind(rep("...", ncol(rem_data_display)),
                        rem_data_display[1,],
                        rep("...", ncol(rem_data_display)),
                        rem_data_display[-1,],
                        rep("...", ncol(rem_data_display)))
rem_data_display <- rbind(rem_data_display,
                          rep(" ", ncol(rem_data_display)))
ft <- flextable(rem_data_display) %>%
  set_table_properties(layout = "fixed", width = 1)
save_as_image(ft, path="pictures/overview/overview_data_fitting.png")
```

```
## [1] "pictures/overview/overview_data_fitting.png"
```

Model Comparison

```
summary(gam_fit)
```

```
##
## Family: binomial
## Link function: logit
##
## Formula:
## y ~ -1 + s(dist_matrix, by = by_matrix) + delta.rec
##
## Parametric coefficients:
##           Estimate Std. Error z value Pr(>|z|)
## delta.rec   0.6236     0.1076   5.793  6.9e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
```

```
##                              edf Ref.df Chi.sq p-value
## s(dist_matrix):by_matrix 2.966  3.633  72.87  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =   -Inf   Deviance explained = -Inf%
## UBRE = 0.26833  Scale est. = 1         n = 1000
```

```r
AIC(gam_fit)
```

```
## [1] 1268.332
```

```r
dat_gam <- remdat

linear_fit_both <-  gam(one ~ delta.dist +
                          delta.rec - 1,
                        family="binomial",
                        data=dat_gam)

linear_rec <- gam(one ~ delta.rec - 1,
                  family="binomial",
                  data=dat_gam)

linear_dist <- gam(one ~ delta.dist - 1,
                   family="binomial",
                   data=dat_gam)

dist_MAT <- cbind(as.numeric(dat_gam$distance),
                  as.numeric(dat_gam$non.distance))
ONE <- cbind(dat_gam$one, -dat_gam$one)
non_linear_dist <- gam(one ~ s(dist_MAT, by=ONE, k=11, bs="bs") - 1,
                       family="binomial",
                       data=dat_gam)
```

```r
AIC(gam_fit)
```

```
## [1] 1268.332
```

```r
AIC(non_linear_dist)
```

```
## [1] 1303.53
```

```r
AIC(linear_fit_both)
```

```
## [1] 1347.209
```

```r
AIC(linear_rec)
```

```
## [1] 1346.776
```

```r
AIC(linear_dist)
```

```
## [1] 1386.387
```

```r
# set.seed(1234)
# BB9 <- BB.single(dim.k = 9)
```

```r
# coefficients(gam_fit)[1]
# GOF_univariate(data = dat_gam, gam.fit = gam_fit,index = 1)[[1]]
```

```r
# coefficients(gam_fit)[2:10]
# GOF_multivariate(data = dat_gam, gam.fit = gam_fit,
#                  index = 2:10, BB.stat = BB9[[2]])[[1]]
# plot(
#   dat_gam$tms,
#   apply(GOF_multivariate(data = dat_gam, gam.fit = gam_fit,
#         index = 2:10, BB.stat = BB9[[2]])[[2]], 1, function(x) crossprod(x,x)),
#   type="l",
#   lwd=2,
#   col="black",
#   xlab="Time Points",
#   ylab="Squared Norm of the Martingale-Residual Process",
#   main="Goodness of Fit of Individual Activity with TVE",
#   ylim=c(0,max(BB9[[1]]))
# )
# for (iter in 1:100){
#   lines(
#     dat_gam$tms,
#     BB9[[1]][seq(1, 2000,
#                  length.out=length(dat_gam$tms)),iter],
#     col="darkgray", lwd=1,
#     lty=3)
# }
```

```r
create_plots <- function(gam_fit, elements, ranges,
                         x_labs, y_labs, titles, subtitle){

  viz_obj <- getViz(gam_fit)

  plots <- list()

  for (element in elements){

    x_lab <- x_labs[element]
    y_lab <- y_labs[element]
    main_title <- titles[element]

    sm <- plot(sm(viz_obj, select=element))
    data <- sm$ggObj$data

    data$upper <- data$y + 1.96 * data$se
    data$lower <- data$y - 1.96 * data$se

    plots[[element]] <- ggplot(data, aes(x = x, y = y)) +
      geom_line(color = "black", linewidth = 0.8) +
      geom_line(aes(y = upper), linetype = "dashed",
                color = "black", linewidth = 0.6) +
      geom_line(aes(y = lower), linetype = "dashed",
                color = "black", linewidth = 0.6) +
      labs(
        x = x_lab,
        y = y_lab,
        title = main_title,
        subtitle = subtitle,
```

```r
      ) +
      theme_minimal(base_size = 14) +
      theme(
        plot.title = element_text(size = 22, face = "bold", hjust = 0.5),
        plot.subtitle = element_text(size = 18, face = "bold", hjust = 0.5),
        axis.title.x = element_text(size = 16),
        axis.title.y = element_text(size = 16),
        axis.text.x = element_text(size = 16),
        axis.text.y = element_text(size = 16),
        axis.text = element_text(size = 13)
      ) +
      ylim(c(ranges[element,]))

  }

  return(plots)

}

plots <- create_plots(gam_fit = gam_fit,
                      elements = c(1),
                      ranges = rbind(c(-0.5,1.2)),
                      x_labs = c("Distance"),
                      y_labs = c("Contribution to Log-Hazard"),
                      titles = c("Non-Linear Effect of Distance"),
                      subtitle = "")

pdf("pictures/overview/distance_overview.pdf", width=6, height=6)
plots[[1]]
dev.off()
```

```
## pdf
##   2
```

```r
exp(coefficients(gam_fit)[1])
```

```
## delta.rec
##  1.865614
```

```r
shifted_remdat_display <- shifted_remdat[x1:x2,c("tms", "s", "r", "weekday",
                                  "non.tms","non.s", "non.r", "non.weekday")]

shifted_remdat_display$tms <- round(shifted_remdat_display$tms, 2)
shifted_remdat_display$non.tms <- round(shifted_remdat_display$non.tms, 2)
shifted_remdat_display$s <- states[shifted_remdat_display$s]
shifted_remdat_display$r <- states[shifted_remdat_display$r]
shifted_remdat_display$non.s <- states[shifted_remdat_display$non.s]
shifted_remdat_display$non.r <- states[shifted_remdat_display$non.r]

shifted_remdat_display <- rbind(rep("...", ncol(shifted_remdat_display)),
                        shifted_remdat_display,
                        rep("...", ncol(shifted_remdat_display)))
colnames(shifted_remdat_display) <- c("Time (Event)",
                              "Origin (Event)", "Destination (Event)",
                              "Weekeday (Event)",
```

```r
                              "Time (Non-Event)", "Origin (Non-Event)",
                              "Destination (Non-Event)", "Weekeday (Non-Event)")

shifted_remdat_display <- rbind(shifted_remdat_display,
                          rep(" ", ncol(shifted_remdat_display)))
rownames(shifted_remdat_display) <- NULL
shifted_remdat_display
```

```
##    Time (Event) Origin (Event) Destination (Event) Weekeday (Event)
## 1           ...            ...                 ...              ...
## 2        381.29           Iowa             Indiana                1
## 3        381.32        Vermont          Washington                1
## 4        381.85       Colorado        Rhode Island                1
## 5        382.91  West Virginia               Texas                1
## 6        383.11       Maryland                Guam                1
## 7        383.29        Arizona            Nebraska                1
## 8        383.43       New York              Hawaii                1
## 9         383.5    Connecticut           Wisconsin                1
## 10       383.98       New York               Texas                1
## 11       384.38        Florida      North Carolina                1
## 12       384.63     Washington             Vermont                1
## 13          ...            ...                 ...              ...
## 14
##    Time (Non-Event) Origin (Non-Event)      Destination (Non-Event)
## 1               ...                ...                          ...
## 2            412.13               Guam                New Hampshire
## 3             307.6              Maine                       Hawaii
## 4            212.71         Washington                         Guam
## 5            458.89         Washington                     Delaware
## 6             77.74              Maine                    Louisiana
## 7             80.81              Maine                   New Jersey
## 8            316.09       Rhode Island                      Georgia
## 9            287.01         New Jersey                West Virginia
## 10            47.28               Iowa United States Virgin Islands
## 11           362.61       South Dakota                      Montana
## 12           434.26        Connecticut                     Colorado
## 13              ...                ...                          ...
## 14
##    Weekeday (Non-Event)
## 1                   ...
## 2                     1
## 3                     1
## 4                     1
## 5                     1
## 6                     0
## 7                     1
## 8                     0
## 9                     0
## 10                    1
## 11                    1
## 12                    0
## 13                  ...
## 14
```

```r
ft <- flextable(shifted_remdat_display) %>%
  set_table_properties(layout = "fixed", width = 1) %>%
  width(j = c(1:8), width = 1.5) %>%
  fontsize(size = 9, part = "all")%>%
  bg(i = 6, j = 4, bg = "#FFE4E1")%>%
  bg(i = 6, j = 8, bg = "#FFE4E1")%>%
  bg(i = 8, j = 4, bg = "#FFE4E1")%>%
  bg(i = 8, j = 8, bg = "#FFE4E1")%>%
  bg(i = 9, j = 4, bg = "#FFE4E1")%>%
  bg(i = 9, j = 8, bg = "#FFE4E1")%>%
  bg(i = 12, j = 4, bg = "#FFE4E1")%>%
  bg(i = 12, j = 8, bg = "#FFE4E1")
save_as_image(ft, path="pictures/overview/overview_shifted_data.png")
```

```
## [1] "pictures/overview/overview_shifted_data.png"
```